



Store at: +2+8°C.

Presentation:

Cod. EZ008

CONT: R1 1 x 40 mL. + R2 1 x 10 mL.

EZ008-SP

CONT: R1 1 x 20 mL. + R2 1 x 5 mL.

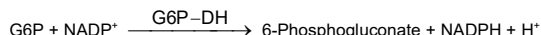
Procedure

Quantitative determination of creatine kinase-MB (CK-MB)

Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

An antibody to the anti CK-M inhibits completely CK-MM and subunit (M) of the CK-MB. The activity of the non-inhibited CK-B subunit is then assayed by the following series of reactions:



The rate of NADPH formation, measured photometrically, is proportional to the catalytic concentration of CK-B present in the sample^{1,2}.

COMPOSICIÓN DE LOS REACTIVOS

R 1 Buffer	Imidazol pH 6,10	125 mmol/L
	Glucose	25 mmol/L
	N-acetylcysteine	25 mmol/L
	NADP ⁺	2,52 mmol/L
	Magnesium acetate	12,5 mmol/L
	EDTA	2 mmol/L
	Hexokinase (HK)	≥6800 U/L
R 2 Substrate	Anti human CK-M antibody (sheep origin) enough to inhibit up to 2000 U/L of CK-MM	
	Imidazol pH 8,90	125 mmol/L
	ADP	15,2 mmol/L
	AMP	25 mmol/L
	di-Adenosine-5- pentaphosphate	103 mmol/L
	Glucosa-6-phosphate dehydrogenase	≥8800 U/L
CONTROL (Optional)	Creatinine phosphate	250 mmol/L
	CK-MB Control: Human liophilized serum 1 x 3 mL.	

PRECAUTIONS

R1/R2: H360D- May damage the unborn child. Restricted to professional users. Follow the precautionary statements given in MSDS and label of the product.

REAGENT PREPARATION AND STABILITY

Working reagent (WR): Mix 1 volume of R2 with 4 volumes of R1. Stability: 1 day at 20-25°C or 2 weeks at 2-8°C.

Signs of Reagent deterioration:

- Presence of particles and turbidity.
- Blank absorbance (A) at 340 nm. ≥ 1.20

All the components of the kit are stable until the expiration date on the label when stored at 2-8°C, protected from light and contamination prevented during their use. Do not use reagents over the expiration date. Do not freeze.

SPECIMEN

Serum free of hemolysis.¹ Samples must be analysed immediately or stability 2 days at 2-8°C or 1 month at -20°C, protected from light. CK-MB activity decreases a 10% after 24 hours at 4°C or 1 hour at 25°C.

MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 340 nm.
- Thermostatic bath at 25°C, 30°C o 37°C (± 0.1°C)
- Matched cuvettes 1.0 cm light path.

General laboratory equipment.

TEST PROCEDURE

- Assay Conditions
 - Wavelength : 340 nm.
 - Cuvette: 1 cm light path.
 - Constant temperature : 25°C / 30°C / 37°C.
- Adjust the instrument to zero with distilled water or air.
- Pipette into a Cuvette^(note 1):

WR (mL)	1.0
Sample (μL)	40

- Mix and incubate for 10 minutes.
- Read the absorbance (A₁) of the sample, start the stopwatch and read absorbance at 5 min. (A₂).
- Calculate the difference of absorbance and the average absorbance difference per minute (ΔA= A₂-A₁).

CALCULATIONS^(Note 2)

$$\Delta A \times 825^* \text{ U/L of CK-B}$$

$$\Delta A \times 1651^* \text{ U/L of CK-MB}$$

Units: One international unit (IU) is the amount of enzyme that transforms 1 μmol of substrate per minute, in standard conditions. The concentration is expressed in units per litre of sample (U/L).

Temperature conversion factors

To correct results to other temperatures multiply by:

Assay temperature	Conversion factor to		
	25°C	30°C	37°C
25°C	1.00	1.53	2.38
30°C	0.65	1.00	1.56
37°C	0.42	0.64	1.00

QUALITY CONTROL

Controls are recommended to monitor the performance of the procedure, CK-MB CONTROL Ref. QC008. If control values are found outside the defined range, check the instrument and reagents for problems.

Controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions.

REFERENCE VALUES

The suspicion of myocardial damage is based on the three following factors.

	25° C.	30° C.	37° C.
CK-MB	>10 U/L.	> 15 U/L.	> 24 U/L.

TOTAL CK

Men, up to 80 U/L. 130 U/L. 195 U/L.

Women up to 70 U/L. 110 U/L. 170 U/L.

Percentage of CK-MB activity in sample:

$$\frac{\text{CK - MB Activity}}{\text{CK Total Activity}} \times 100 : 6-25 \% \text{ CK-MB Activity in the sample}$$

(These values are for orientation purpose).

It is suggested that each laboratory establish its own reference range.

CLINICAL SIGNIFICANCE

CK-MB is an enzyme formed by the association of two subunits from muscle (M) and nerve cells (B). CK-MB is usually present in serum at low concentration; it is increases after an acute infarct of myocardium and later descends at normal levels. Also is increased, rarely, in skeletal muscle damage^{3,6}.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

Measuring Range:

From detection limit of 15 U/L. (on Cobas Mira) to linearity limit of 600 U/L., under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample 1/10 with NaCl 9 g/L. and multiply result by 10.

Precision:

	Intra-assay n= 20		Inter-assay n= 20	
Mean (U/L)	24,95	66	25	74
CV (%)	10,36	4,59	9,80	2,62

Sensitivity: 6 U/L (on Cobas Mira).

Accuracy: Results obtained reagents did not show systematic differences when compared with other commercial reagents.

Correlation coefficient (r): 0.99

Regression Equation: y=1.0183x + 0.308

The results of the performance characteristics depend on the analyzer used.

INTERFERING SUBSTANCES

- No interferences were observed with glucose until 7 g/L., haemoglobin until 6 g/L. and triglycerides 8 mmol/L.

- A list of drugs and other interfering substances with CK determination has been reported by Young et. al.^{3,4}.

NOTES

- Use clean disposable pipette tips for its dispensation.
- Formulation to reach constant:

$$\Delta A/5 \text{ min} \times 825^* \text{ or } 1651^* = \text{U/L CK}$$

* Tv x 1000 ε x LP x Sv	Tv= Total volume in mL ε NADPH = 6.22 at 340 nm LP= Light path Sv= Sample volume in mL

BIBLIOGRAPHY

- Abbot B et al. Creatinine kinase. Kaplan A et al. Clin Chem The C.V. Mosby Co. St Louis. Toronto. Princeton 1984: 1112-116.
- Gerhardt W. et al. Creatine kinase B-Subunit activity in serum after immunoinhibition of M-Subunit activity. Clin Chem 1979;(25/7): 1274-1280.
- Young DS. Effects of drugs on Clinical Lab. Tests, 4th ed AACC Press, 1995.
- Young DS. Effects of disease on Clinical Lab. Tests, 4th ed AACC 2001.
- Burtis A et al. Tietz Textbook of Clinical Chemistry, 3rd ed AACC 1999.
- Tietz N W et al. Clinical Guide to Laboratory Tests, 3rd ed AACC 1995.
- Mathieu M. et coll. Recommendation por la mesure de la concentration catalytique de la créatine kinase dans le serum humain. Ann. Biol. Clin.,40 (1482), 87.
- Neumeier, D., Prellwitz, W., Würzburg, U. Et coll. Determination of creatine kinase isoenzyme MB activity in serum using immunological inhibition of Creatine kinase M subunit activity. Activity kinetics and diagnostics significance in myocardial infarction, Clin. Chim. Acta, 73, (1976), 445.



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